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SEMESTER IV

AD23411 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	**HEALTHCARE DATA ANALYTICS FOR DISEASE PATTERN INDENTIFICATION**
DATE OF SUBMISSION	
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Signature of Faculty In-charge **Signature of HoD**

Internal Examiner **External Examiner**

INTRODUCTION

Data Analytics plays a vital role in the healthcare sector by enabling the analysis of large volumes of patient data to extract meaningful insights. With the increasing availability of healthcare datasets, it has become possible to identify patterns in diseases, understand risk factors, and improve early diagnosis.

One of the major challenges in healthcare is identifying diseases at an early stage and understanding the factors that contribute to their occurrence. Traditional methods rely on manual analysis, which is often time-consuming and less accurate when dealing with complex datasets.

Disease Pattern Identification is a data-driven approach that helps in analysing patient attributes such as age, medical history, and clinical measurements to detect patterns associated with diseases. By studying historical healthcare data, it becomes possible to predict whether a patient is likely to develop a disease.

In this project, healthcare data is analyzed using data analytics techniques. The system processes patient-related attributes and identifies relationships between different health indicators and disease occurrence.

Exploratory Data Analysis (EDA) and visualization techniques such as histograms, bar charts, and heatmaps are used to understand data distribution and correlations. A machine learning model (Random Forest Classifier) is used to predict disease presence.

Overall, this project demonstrates how data analytics can help in disease prediction, improve healthcare decision-making, and assist in early diagnosis.

PROBLEM STATEMENT

In the healthcare domain, identifying disease patterns and risk factors is a major challenge due to the complexity and volume of medical data. Manual analysis is not sufficient to detect hidden relationships between patient attributes and diseases.

Healthcare datasets often contain both numerical and categorical data, missing values, and inconsistencies, making analysis difficult. Traditional approaches fail to consider multiple influencing factors such as age, cholesterol levels, blood pressure, and lifestyle simultaneously.

The problem addressed in this project is the need for an efficient system that can analyze healthcare data, identify disease patterns, and predict the presence of diseases accurately.

The system should:

- Handle missing and inconsistent data
- Analyse relationships between health indicators
- Identify key risk factors
- Provide accurate disease predictions

- Provides easy-to-understand results
- Supports early disease detection
- Works well on new/unseen data

GOAL

The primary goal of this project is to develop a Healthcare Data Analytics System that can analyze patient data and identify patterns related to disease occurrence. The system is designed to process various health-related attributes and generate meaningful insights along with disease predictions.

One of the key objectives is to perform effective data preprocessing, including handling missing values, encoding categorical variables, and cleaning the dataset. This ensures that the data is accurate and suitable for analysis.

Another important goal is to apply exploratory data analysis techniques to understand patterns in healthcare data and identify factors influencing diseases. By analyzing attributes such as age, cholesterol, and blood pressure, the system provides valuable insights into disease risk.

The project also aims to build a predictive model (Random Forest Classifier) that can classify patients into disease and non-disease categories. The model should be accurate, reliable, and capable of handling real-world healthcare data.

In addition, the system focuses on visualization by presenting results through charts and graphs. These visualizations help in understanding disease distribution and identifying key risk factors.

Furthermore, the project aims to provide a simple system where users can input patient details and obtain predictions, making it useful for healthcare analysis.

In conclusion, the goal of this project is to develop an efficient, scalable, and data-driven solution that helps in disease prediction, early diagnosis, and improved healthcare decision-making.

THEORETICAL BACKGROUND

Healthcare Data Analytics for Disease Pattern Identification is an important application of Data Analytics that focuses on identifying patterns in disease occurrence using patient data. It combines concepts from data preprocessing, statistical analysis, and machine learning to analyze health indicators and detect relationships between them. This approach helps in predicting diseases, understanding risk factors.

1. Data Analytics

Data analytics involves analyzing datasets to extract useful information. In this project:

- Descriptive Analytics → Understand data distribution
- Predictive Analytics → Predict disease occurrence

2. Healthcare Data

Healthcare datasets include:

- Age
- Blood pressure
- Cholesterol levels
- Heart rate
- Diagnosis (disease presence)

3.Data Preprocessing

Data preprocessing is essential for preparing raw data for analysis. It includes:

- Handling missing values
- Encoding categorical variables
- Feature scaling and normalization

These steps improve data quality and model performance.

4.Disease Prediction Techniques

Disease prediction can be performed using:

- Rule-based methods
- Statistical analysis
- Machine learning models (e.g., Random Forest, Logistic Regression)

In this project, predictive techniques are used to classify patients into disease and non-disease categories.

5. Feature Importance

Feature importance helps identify which factors most influence disease occurrence. Common influencing factors include:

- Age
- Blood pressure
- Cholesterol levels
- Heart rate and other medical attributes

Understanding these factors helps in early diagnosis and better healthcare decision-making.

6.Data Visualization

Visualization helps present insights clearly. Common techniques include:

• Bar charts (disease vs non-disease comparison)
• Histograms (distribution of age and health parameters)
• Heatmaps (correlation between features)

These visual tools improve interpretability and support better healthcare decision-making

7.Statistical Analysis

Statistical methods are used to summarize and analyze healthcare data:

• Frequency distribution
• Mean and variance
• Correlation analysis

These techniques help in understanding relationships between health indicators and disease occurrence.

METHODOLOGY

The methodology of this project follows a structured data analytics pipeline to process, analyze, and predict disease occurrence. The workflow consists of multiple stages, starting from data collection to final prediction and insight generation. Each step is designed to ensure accurate analysis and reliable results.

1. Data Collection

The first step involves collecting relevant healthcare data for analysis. The dataset can be obtained from publicly available sources such as Kaggle or the UCI Machine Learning Repository (e.g., Heart Disease Dataset).

The dataset typically includes patient-related attributes such as:

• Patient ID
• Age
• Gender
• Blood pressure
• Cholesterol levels
• Heart rate
• Other medical attributes
• Disease (target variable)

This data forms the foundation for analyzing health patterns and predicting diseases.

2. Data Cleaning and Preprocessing

Raw healthcare data may contain inconsistencies, missing values, and irrelevant information. Therefore, preprocessing is essential to improve data quality.

The following steps are performed:

• Handling missing or null values
• Removing unnecessary columns (e.g., ID)
• Converting categorical values into numerical format (encoding)
• Standardizing data formats for consistency

These steps ensure that the dataset is clean, structured, and ready for analysis.

3.Data Transformation

After preprocessing, the data is transformed into a suitable format for analysis and modeling. Key transformation steps include:

• Encoding categorical variables
• Converting target variable (Disease: Yes/No → 1/0)
• Feature scaling (if required)
• Splitting dataset into input features (X) and target variable (y)

This step ensures that all variables are in a consistent format for further processing.

4.Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand patterns and relationships within the dataset.

The following analyses are conducted:

• Distribution of disease vs non-disease patients
• Relationship between age and disease
• Impact of cholesterol and blood pressure on disease
• Analysis of different health indicators

EDA helps in identifying key factors influencing disease occurrence and provides valuable insights.

5.Model Development

A predictive model is developed to classify patients into disease and non-disease categories. In this project, a classification algorithm such as:

• Random Forest Classifier

is used to build the model. Steps involved:

• Splitting data into training and testing sets
• Training the model using historical data
• Generating predictions on test data

This step enables the system to predict whether a patient is likely to have a disease.

6.Model Evaluation

The performance of the model is evaluated using standard metrics to ensure accuracy and reliability.

Evaluation techniques include:

• Accuracy Score
• Precision and Recall
• Confusion Matrix

These metrics help in measuring how well the model predicts disease occurrence.

7.Data Visualization

Visualization is used to present insights in an understandable format. The following visualizations are generated:

• Bar Charts: Compare disease and non-disease patients
• Histograms: Show distribution of age and health parameters
• Correlation Heatmaps: Identify relationships between features
• Feature Importance Graphs: Highlight key factors affecting disease

These visual tools help in interpreting results and understanding health patterns.

8.Insight Generation

Based on the analysis and model results, meaningful insights are derived, such as:

• Identification of high-risk patients
• Key factors influencing disease (e.g., cholesterol, blood pressure)
• Health patterns associated with disease occurrence

These insights help in early diagnosis and better healthcare decisions.

9.User Interaction (Optional Web Interface)

To enhance usability, a simple interface can be developed where users can:

• Input patient details
• Get instant disease prediction results
• View visual insights and reports

SAMPLE DATA

id	age	sex	data set	cp	trest	chol	fbs	restecg	Thalch	Ex ang	Old peak	slope	ca
1	63	M	cleveland	typical angina	145	233	T	lv hypertrophy	150	F	2.3	downsloping	0
2	67	M	cleveland	asymptomatic	160	286	F	lv hypertrophy	108	T	1.5	flat	3
3	67	M	cleveland	asymptomatic	120	229	F	lv hypertrophy	129	T	2.6	flat	2
4	37	M	cleveland	non-anginal	130	250	F	normal	187	F	3.5	downsloping	0
5	41	F	cleveland	atypical angina	130	204	F	lv hypertrophy	172	F	1.4	upsloping	0
6	56	M	cleveland	atypical angina	120	236	F	normal	178	F	0.8	upsloping	0
7	62	F	cleveland	asymptomatic	140	268	F	lv hypertrophy	160	F	3.6	downsloping	2
8	57	F	cleveland	asymptomatic	120	354	F	normal	163	T	0.6	upsloping	0
9	63	M	cleveland	asymptomatic	130	254	F	lv hypertrophy	147	F	1.4	flat	1

Graphs and Charts

The analysis of healthcare data is supported using various visualizations that help in understanding disease patterns, identifying key risk factors, and extracting meaningful insights. These graphical representations enhance interpretability and assist in making data-driven healthcare decisions.

- **Disease Distribution Chart (count plot):**

This chart represents the overall distribution of patients with and without disease. It provides a clear overview of disease occurrence within the dataset.
In this project, both categories (disease and non-disease) are observed, helping in understanding the proportion of affected individuals and the need for early detection.

- **Age Distribution (Histogram):**

This visualization shows the distribution of patients across different age groups.
The graph indicates that certain age groups are more prone to disease, highlighting age as an important factor influencing disease occurrence.

- **Health Indicators vs Disease (Analysis):**

This analysis shows the relationship between health parameters such as cholesterol and blood pressure with disease.
It can be observed that abnormal values in these parameters increase the likelihood of disease,

indicating their significance as risk factors.

- **Correlation Heatmap:**

This graph shows the relationship between different health attributes and disease.
The visualization helps in identifying which features have strong positive or negative correlation with disease.

- **Feature Importance Graph:**

This chart highlights the most important features influencing disease prediction.
The model shows that factors like age, cholesterol, and blood pressure play a major role in identifying disease risk

IMPLEMENTATION CODE AND DATASET USED

Category	Description
Dataset	Heart Disease UCI
Domain	Healthcare Analytics
Objective	Disease Prediction
Data Type	Tabular
Features	Age, BP, Cholesterol, etc.
Target	Disease (0/1)
Model	Random Forest
Libraries	pandas, numpy, sklearn, matplotlib, seaborn
Metrics	Accuracy, Confusion Matrix, Classification Report

Dataset Description

The dataset used in this project contains patient-related healthcare information. It includes medical attributes, clinical measurements, and diagnosis details. Each record represents an individual patient and indicates whether the patient has a disease or not.

The dataset is structured in tabular format, making it suitable for classification problems. It contains both numerical and categorical features, which require preprocessing before analysis.

Each record includes:

• Patient-related details (age, gender)
• Medical attributes (blood pressure, cholesterol, heart rate)
• Clinical measurements and test results
• Target variable indicating disease presence

This dataset is primarily used for:

• Identifying patterns in disease occurrence
• Analyzing factors influencing diseases
• Building predictive models for disease classification

PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix,
accuracy_score


sns.set(style="whitegrid")


df = pd.read_csv(r"C:\Users\Canara\Downloads\heart_disease_uci.csv")

print("\nFirst 5 Rows:\n", df.head())


df.replace('?', np.nan, inplace=True)


num_cols = df.select_dtypes(include=['float64', 'int64']).columns
cat_cols = df.select_dtypes(include=['object']).columns


df[num_cols] = df[num_cols].fillna(df[num_cols].mean())
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])


df = pd.get_dummies(df, drop_first=True)

print("\nMissing Values After Cleaning:\n", df.isnull().sum())


df['target'] = df['num'].apply(lambda x: 1 if x > 0 else 0)


print("\nStatistical Summary:\n", df.describe())

plt.figure()
```

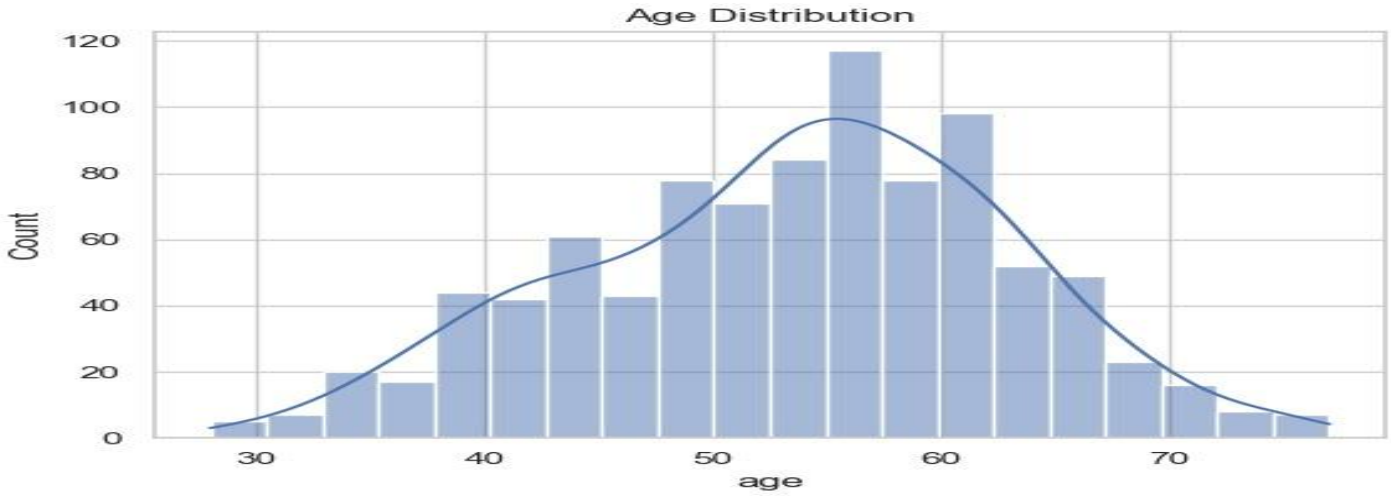
```
sns.histplot(df['age'], bins=20, kde=True)
plt.title("Age Distribution")
plt.show()
plt.figure()
sns.countplot(x='target', data=df)
plt.title("Disease Distribution (0 = No, 1 = Yes)")
plt.show()
corr = df.corr()
plt.figure(figsize=(12, 8))
sns.heatmap(corr, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
target_corr = corr['target'].sort_values(ascending=False)
print("\nCorrelation with Disease:\n", target_corr)
plt.figure()
target_corr.drop('target').head(10).plot(kind='bar')
plt.title("Top Risk Factors")
plt.ylabel("Correlation Value")
plt.show()
X = df.drop(['target', 'num', 'id'], axis=1)
y = df['target']

print("\nFeature Shape:", X.shape)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nModel Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

importances = model.feature_importances_
features = X.columns

feature_importance = pd.Series(importances, index=features)
feature_importance = feature_importance.sort_values(ascending=False)

print("\nFeature Importance:\n", feature_importance.head(10))

plt.figure()
feature_importance.head(10).plot(kind='barh')
plt.title("Top Features Influencing Disease")
```

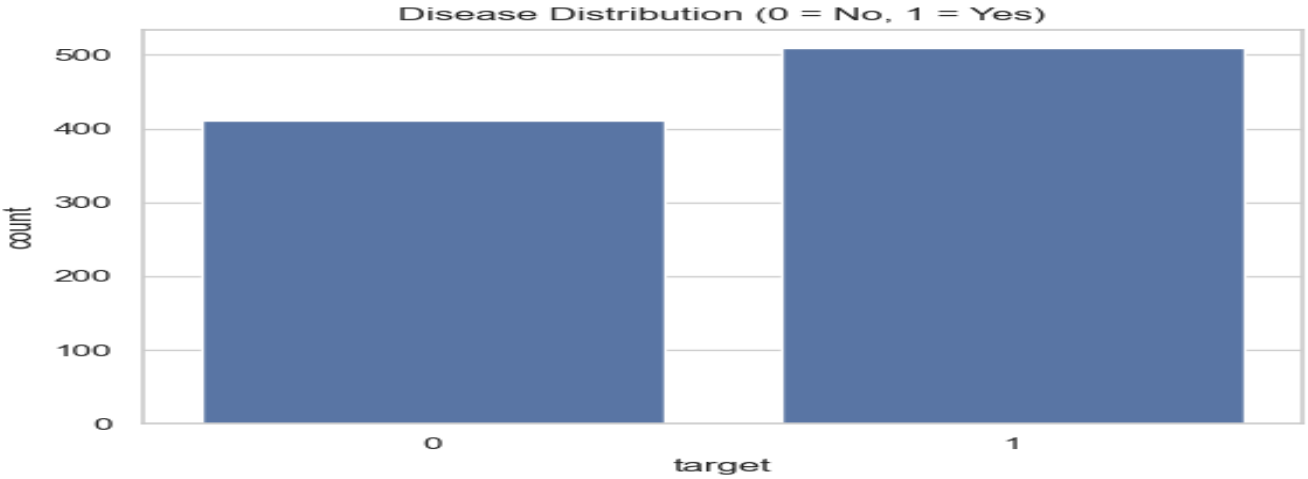
```
plt.xlabel("Importance Score")
plt.show()
```

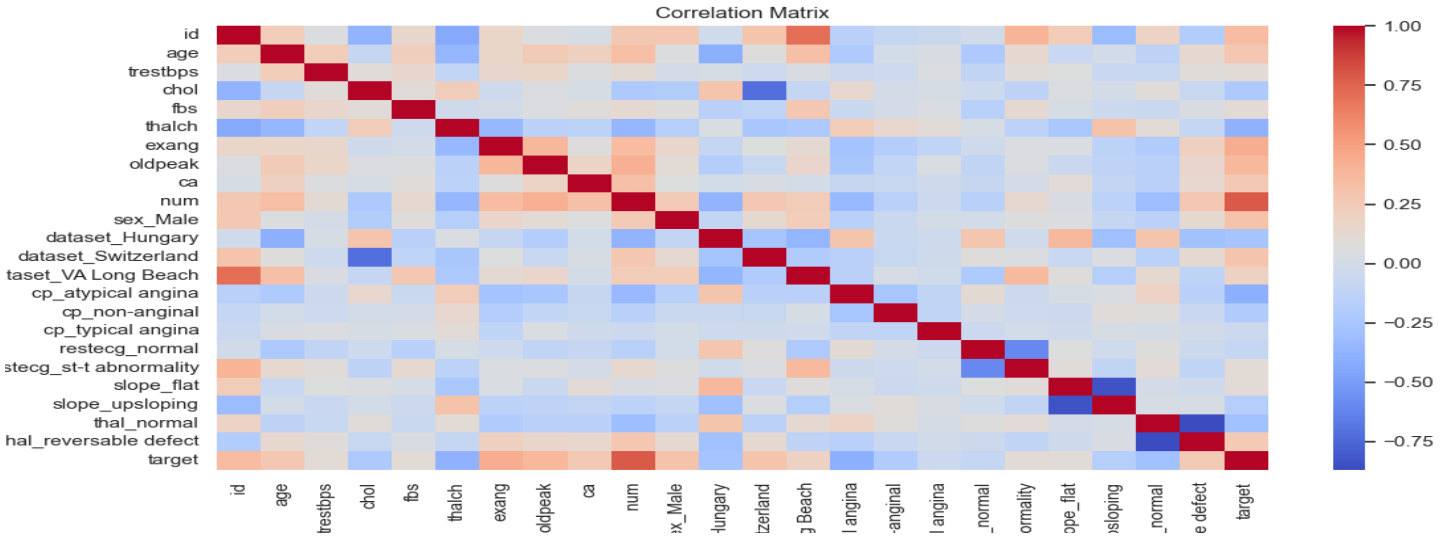
OUTPUTS

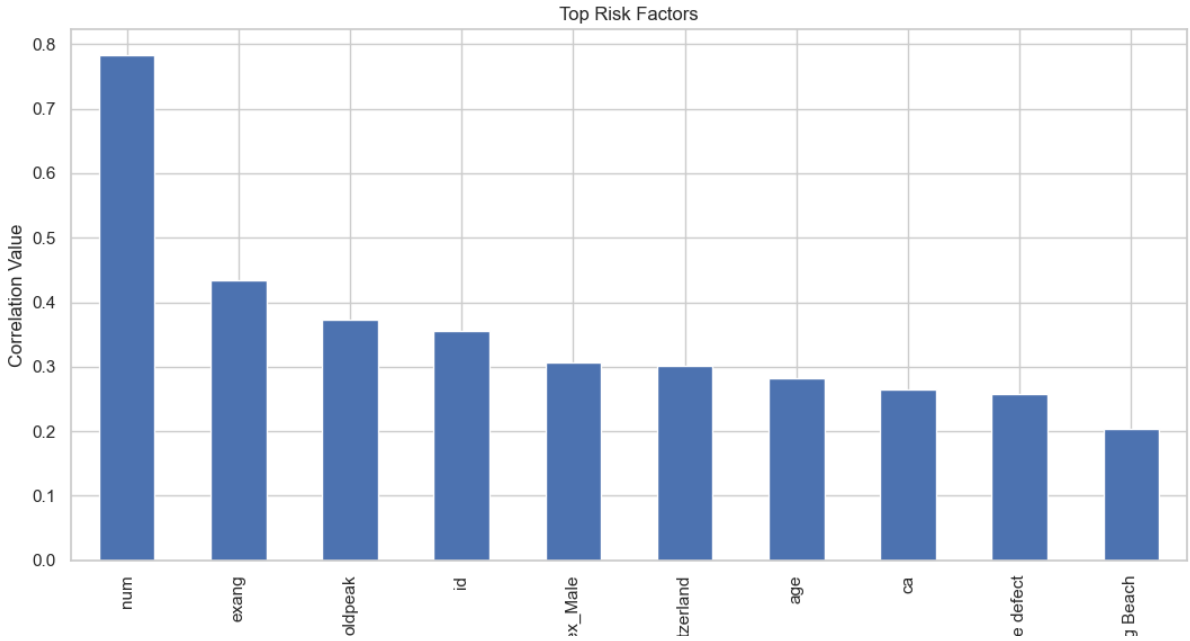
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

PS C:\Users\Canara\OneDrive\Desktop\DA lab> & "C:\Program Files\Python313\python.exe" "c:/Users/Canara/OneDrive/Desktop/DA lab/Mini_project.py"

First 5 Rows:
    id  age     sex    dataset              cp  trestbps   chol    fbs        restecg  thalch  exang  oldpeak       slope   ca              thal  num
0   1   63    Male  Cleveland   typical angina     145.0  233.0   True  lv hypertrophy   150.0  False      2.3  downsloping  0.0      fixed defect    0
1   2   67    Male  Cleveland     asymptomatic     160.0  286.0  False  lv hypertrophy   108.0   True      1.5         flat  3.0            normal    2
2   3   67    Male  Cleveland     asymptomatic     120.0  229.0  False  lv hypertrophy   129.0   True      2.6         flat  2.0  reversable defect    1
3   4   37    Male  Cleveland      non-anginal     130.0  250.0  False          normal   187.0  False      3.5  downsloping  0.0            normal    0
4   5   41  Female  Cleveland  atypical angina     130.0  204.0  False  lv hypertrophy   172.0  False      1.4    upsloping  0.0            normal    0
c:\Users\Canara\OneDrive\Desktop\DA lab\Mini_project.py:16: FutureWarning: Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated and will
change in a future version. Call result.infer_objects(copy=False) instead. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting
', True)`
  df[col] = df[col].fillna(df[col].mode()[0])

Missing Values After Cleaning:
 id                        0
age                       0
trestbps                  0
chol                      0
fbs                       0
thalch                    0
exang                     0
oldpeak                   0
ca                        0
num                       0
sex_Male                  0
dataset_Hungary           0
dataset_Switzerland       0
dataset_VA Long Beach     0
cp_atypical angina        0
cp_non-anginal            0
cp_typical angina         0
restecg_normal            0
```

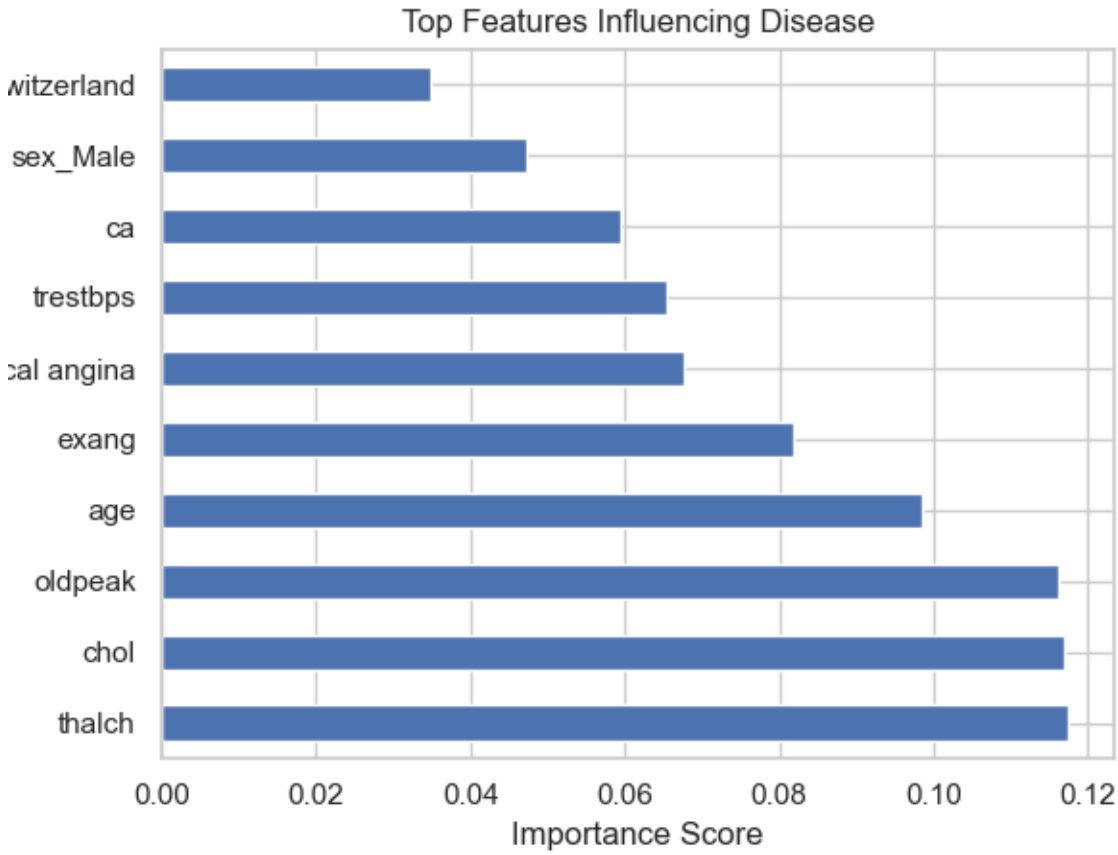
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Canara\OneDrive\Desktop\DA lab> & "C:\Program Files\Python313\python.exe" "c:/Users/Canara/OneDrive/Des
restecg_st-t abnormality    0
slope_flat                  0
slope_upsloping             0
thal_normal                 0
thal_reversable defect      0
dtype: int64

Statistical Summary:
                id         age    trestbps        chol      thalch     oldpeak          ca         num      target
count  920.000000  920.000000  920.000000  920.000000  920.000000  920.000000  920.000000  920.000000  920.000000
mean   460.500000   53.510870  132.132404  199.130337  137.545665    0.878788    0.676375    0.995652    0.553261
std    265.725422    9.424685   18.443895  108.957634   25.138494    1.053774    0.541667    1.142693    0.497426
min      1.000000   28.000000    0.000000    0.000000   60.000000   -2.600000    0.000000    0.000000    0.000000
25%    230.750000   47.000000  120.000000  177.750000  120.000000    0.000000    0.676375    0.000000    0.000000
50%    460.500000   54.000000  130.000000  221.000000  138.000000    0.800000    0.676375    1.000000    1.000000
75%    690.250000   60.000000  140.000000  267.000000  156.000000    1.500000    0.676375    2.000000    1.000000
max    920.000000   77.000000  200.000000  603.000000  202.000000    6.200000    3.000000    4.000000    1.000000

Correlation with Disease:
 target                      1.000000
num                          0.783387
exang                        0.433605
oldpeak                      0.372890
id                           0.356086
sex_Male                     0.307284
dataset_Switzerland          0.301614
age                          0.282700
ca                           0.264936
thal_reversable defect       0.257029
dataset_VA Long Beach        0.203267
fbs                          0.108071
restecg_st-t abnormality     0.104750
Feature Shape: (920, 21)

Model Accuracy: 0.8532608695652174

Confusion Matrix:
 [[66  9]
 [18 91]]

Classification Report:
              precision    recall  f1-score   support

           0       0.79      0.88      0.83        75
           1       0.91      0.83      0.87       109

    accuracy                           0.85       184
   macro avg       0.85      0.86      0.85       184
weighted avg       0.86      0.85      0.85       184


Feature Importance:
 thalch                 0.117506
chol                   0.117003
oldpeak                0.116050
age                    0.098564
exang                  0.081935
cp_atypical angina     0.067581
trestbps               0.065408
ca                     0.059335
sex_Male               0.047285
dataset_Switzerland    0.034712
dtype: float64
```


Age Distribution
Count
120
100
80
60
40
20
0
30
40
50
60
70
age


Disease Distribution (0 = No, 1 = Yes)
count
500
400
300
200
100
0
0
1
target


Correlation Matrix
id
age
trestbps
chol
fbs
thalch
exang
oldpeak
ca
num
sex_Male
dataset_Hungary
dataset_Switzerland
taset_VA Long Beach
cp_atypical angina
cp_non-anginal
cp_typical angina
restecg_normal
stecg_st-t abnormality
slope_flat
slope_upsloping
thal_normal
hal_reversable defect
target
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Top Risk Factors
Correlation Value
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
num
exang
oldpeak
id
age
ca


Top Features Influencing Disease
sex_Male
ca
trestbps
exang
age
oldpeak
chol
thalch
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance Score

RESULTS AND FUTURE ENHANCEMENT

RESULTS

The Healthcare Data Analytics System was successfully implemented to analyze patient data and predict the presence of diseases. The system effectively processed structured healthcare data through preprocessing steps such as handling missing values, encoding categorical variables, and transforming features, ensuring high data quality for analysis.

The disease distribution analysis shows that the dataset contains both diseased and non-diseased patients, helping in understanding the overall health pattern. The visualization clearly represents this distribution, highlighting the importance of early detection.

The exploratory data analysis provided valuable insights into health patterns. The age distribution analysis indicates that certain age groups are more prone to diseases, emphasizing age as an important risk factor.

The correlation analysis reveals relationships between various health indicators and disease occurrence. Factors such as cholesterol levels, blood pressure, and other medical attributes show strong influence on disease prediction.

The feature importance analysis further identifies key risk factors contributing to disease. These insights help in understanding which health parameters play a major role in disease occurrence.

The predictive model (Random Forest Classifier) demonstrated good performance in classifying patients into disease and non-disease categories. The evaluation metrics, including accuracy, confusion matrix, and classification report, indicate that the model is effective in identifying disease patterns.

Overall, the system successfully combines data analysis, visualization, and predictive modeling to generate meaningful insights. It helps in identifying high-risk patients and supports better healthcare decision-making and early diagnosis.

FUTURE ENHANCEMENTS

• Increase the dataset size by incorporating more real-world healthcare data to improve model accuracy

• Apply advanced machine learning algorithms such as XGBoost or Neural Networks for better prediction performance

• Perform hyperparameter tuning to optimize the model and reduce overfitting

• Include additional health parameters such as lifestyle habits, medical history, and genetic data

• Implement real-time data integration for continuous monitoring and prediction

• Develop a user-friendly web or mobile application for easy disease prediction

• Enhance data visualization with interactive dashboards and advanced analytics

• Integrate the system with hospital databases for practical real-world usage

Git Hub Link of the project along with the report	https://github.com/sbhavatharini0110-web/Data_Analytics_miniproject

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